Movie quality was tested under the 'Cinema' preset of the projectors. We used scenes shot in daylight and at night to gauge the performance, especially the details they produced (such as the bark of a tree, skin tones and whether a scene was overly dark, which results in to detail loss). In this test, the Toshiba TDP-S20 was better than the rest. It produced night scenes precisely and with good detail reproduction. Sony's was the only LCD projector to produce an image as good as Toshiba's.

Another worthy mention here is the Epson EMP-S1H-the one with the efficient lamp and the large projection size. It not only produced good details and skin tones but its bigger display allowed for a truly cinematic experience. That said, you will need to sit at least 20 to 25 feet away from the screen to get rid of the screen door effect: the screen door effect is the grid visible on a screen because of the space between two pixels. Remember that this is an SVGA projector, so it is spreading only 320,000 pixels over a 330-inch diagonal screen.

Toshiba TDP-S20

The HP vp6110 was a letdown in this test. The image produced at full zoom was no more than 60 to 65 inches and dark enough to blank out the night scene! Even the daylight scenes appeared dull and details were completely lost. This means that the presets on the HP vp6110 are no good.

DisplayMate Video Edition

The DisplayMate Video Edition test uses a set of visually brilliant slides, and the primary reason to use it here was to check the display of the projectors under extreme conditions.

When we looked for geometric accuracy and focusing, Acer, BenQ and HP did extremely well. These three, with their default settings, gave accurate geometry in horizontal and vertical bar resolutions with no visible moiré. The projected image was accurately focused throughout. The Toshiba TDP-S20 suffered in the focus matrix test and displayed prominent barrel distortion aberration. The problem was not specific to the piece tested—a replacement unit too showcased similar issues. The Epson was another one with poor results in the focus matrix test: images in corners were completely out of focus and remained that way even when we tried different modes.

Given its performance, the Acer PD113—a DLP-based projector—topped the charts in this category leaving behind Sony

When we looked for streaking and ghosting, purity of colour and level shift, the BenQ PB6100 and the HP vp6110 shared the top spot. With these models, there was no ghosting or visible white or black level shift at all. The units even managed to accurately differentiate between 16 grey levels of varying intensity. The Canon and Toshiba units had problems of white and black level shift, visible in the form of a red and a blue tinge where the bright and the dark areas met.

We also checked for screen uniformity and flicker. The Toshiba TDP-S20 and the Sharp PG-B10S displayed problems with screen uniformity: the projected screen was clearly dark on either side of the centre. The BenQ, HP and Infocus returned commendable results in this test. The Sharp unit showed a slight flicker. In the dark screen test, Sony and Sharp appeared brighter, and therefore, a distinct dark grey image was formed on the screen; the others did well too, projecting dark enough screens.

Total Cost of Lamp

The cost of a lamp is an important factor to be considered while purchasing a projector. Most projectors have a lamp life of 2,000 hours. A few, such as the Infocus, have a lamp life of 4,000 hours while the Sony and Toshiba units have a 3,000-hour lamp life. The cost of the Infocus lamp is high, at Rs 32,000. But if we calculate the per-hour cost, it comes to Rs 8, which is lower than the Rs 12.5 of the HP, Canon and Epson, whose lamps retail for Rs 25,000 and last 2,000 hours. The most affordable lamp in terms of per-hour rate, was that of the Sony. The lamp cost is Rs 17,990, the lowest in this category, and the per-hour cost of using the lamp comes out to Rs 6, which is half that of the Toshiba, HP, Canon or Epson.

And The Winners Are...

Given its performance, the Acer PD113—a DLP-based projector—topped the charts in this category, leaving behind Sony by a percentage point and a half. The Acer sported a good set of features, prime amongst them being a healthy 1,600 lumen brightness and a zoom ratio of 1.2:1. It also had a high image display size of

Jargon Buster

Anamorphic Lens: A lens that can project an aspect ratio different from the actual image ratio is called an Anamorphic Lens. Using such a lens, it is possible to project 16:9 aspect ratio images using a native 4:3 aspect ratio projector

ANSI Contrast Ratio: The ratio of the brightest white to the darkest black. Display devices supporting higher contrast ratio will be able to show much finer colour range and detail. ANSI Contrast is a method of measuring contrast by using 16 alternating white and black rectangles. The average output of light from white and black is divided to get the ANSI contrast ratio

Composite Video Signal: A signal which is a combination of colour information and horizontal and vertical sync signals is called a Composite Signal. Television signals are composite signals

Dichroic: A lens, typically used in projectors, separates white information from the red, green and blue light. Such a lens is called Dichroic. It does this by reflecting/refracting particular wavelengths of light

Keystone: When a projected image does not appear rectangular on the screen, the glitch is called Keystone. It usually happens when the projector is not perpendicular to the screen. It can be corrected by either mechanically shifting the lens or digitally correcting the image to make it look rectangular

Lens Shift: The ability to move the lens horizontally or vertically is called Lens Shift. Some entry-level projectors have a manual lens shift where the user can shift the lens using a small joystick located beside the display lens